

Set-Based Uncertainty Propagation in Orbital Dynamics: Automatic Domain Splitting over Zonotopes and Polynomial Zonotopes

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Abstract

Propagating an initial-condition uncertainty set through nonlinear orbital dynamics is a core problem in space situational awareness, conjunction assessment, and orbit determination. Linear (covariance) propagation is cheap but loses validity as nonlinearity grows; Monte Carlo is general but expensive and noisy in the tails. Differential-algebra (DA) methods occupy the middle ground by propagating a truncated Taylor expansion of the flow map, and *automatic domain splitting* (ADS) restores accuracy by subdividing the domain wherever the expansion stops converging. We observe that the object ADS manipulates—a Taylor flow map over a bounded domain of expansion variables—is exactly a *polynomial zonotope*, the set representation developed independently in the reachability-analysis community. Building on this correspondence, we study three usage patterns for combining ADS with zonotopic representations on the planar two-body problem: (i) *oriented* domains, in which the factor frame is a general parallelotope rather than an axis-aligned box, reducing the leaf count for correlated uncertainty; (ii) *flow-aligned* domains, in which the orientation of an ellipsoidal initial set is chosen from the singular vectors of the state-transition matrix; and (iii) *curved* initial sets, in which the initial geometry is itself a polynomial zonotope, as arises when a Gaussian in orbital elements is expressed in Cartesian state. All results are validated against 10^4 Monte Carlo samples. The combined approach reproduces the propagated set to the requested tolerance while using far fewer sub-domains than an axis-aligned box, and the curved representation captures non-Gaussian initial uncertainty that no affine set can represent.

1 Introduction

A recurring task in astrodynamics is to map a set of possible initial states through the equations of motion and characterise the resulting set of possible final states. The set may encode navigation uncertainty, manoeuvre dispersions, an admissible region for a too-short-arc detection, or the family of states consistent with an observation. Once the set is propagated, downstream questions follow: does it intersect a keep-out volume, what is the collision probability, where should a follow-up observation point.

Three families of methods dominate practice [11]. *Linear covariance* propagation maps a Gaussian through the state-transition matrix; it is inexpensive but assumes the dynamics are well approximated by their linearisation over the support of the distribution, which fails for large sets or long horizons [7]. *Monte Carlo* samples the initial distribution and propagates each sample; it is asymptotically exact and trivially parallel but converges slowly and is noisy precisely in the low-probability tails that matter for risk. Between the two, *differential-algebra* (DA) methods propagate a truncated Taylor expansion of the flow map about a reference, yielding a polynomial that can be evaluated anywhere in the domain [3–5, 13]. The polynomial

is accurate only while the Taylor series converges over the domain; *automatic domain splitting* (ADS) recovers accuracy for large sets by subdividing the domain whenever the truncation error of the current expansion exceeds a threshold, so that the propagated set is described by a union of polynomial pieces, each valid on its sub-domain [10, 14].

In parallel, the formal-verification and control communities developed *set propagation* techniques for reachability analysis, with a hierarchy of set representations: intervals, *zonotopes* [6], *polynomial zonotopes* [1, 8], and their constrained generalisations [9, 12]; see Althoff et al. [2] for a survey. The central observation of this paper is that the DA flow map propagated by ADS is a polynomial zonotope (Section 3). This identification makes the machinery of zonotopic representations—in particular the freedom to choose the geometry of the factor domain—available to orbital-uncertainty propagation, and conversely gives the reachability representations an adaptive, error-driven subdivision strategy in ADS.

We study three concrete usage patterns and their effect on cost and accuracy, using the planar two-body problem as a controlled test bed:

1. **Oriented domains** (Section 5): replacing the axis-aligned box of factors by a general parallelotope (a zonotope) wraps a correlated initial set more tightly and reduces the number of sub-domains.
2. **Flow-aligned domains** (Section 6): for an ellipsoidal (covariance) initial set, the covering parallelotope is free to orient; choosing its orientation from the singular value decomposition of the state-transition matrix conditions the propagated set and minimises subdivision.
3. **Curved initial sets** (Section 7): when the uncertainty is Gaussian in one coordinate system and expressed in another through a nonlinear map (e.g. orbital elements to Cartesian state), the initial set is itself a polynomial zonotope. Representing it as such captures non-Gaussian structure that an affine set cannot.

Every claim is checked against 10^4 Monte Carlo samples (Section 8). The paper concerns the *usage* of these representations for orbital dynamics; the underlying numerics are standard DA and ADS, and the implementation is incidental to the discussion.

2 Background

2.1 Flow map and uncertainty propagation

Consider an autonomous ordinary differential equation

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n, \quad (1)$$

with flow $\varphi_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$, so that $\varphi_t(\mathbf{x}_0)$ is the solution at time t from initial state $\mathbf{x}_0$. Given an initial set $\mathcal{X}_0 \subset \mathbb{R}^n$, the propagated set is its image under the flow,

$$\mathcal{X}_t = \varphi_t(\mathcal{X}_0) = \{ \varphi_t(\mathbf{x}_0) : \mathbf{x}_0 \in \mathcal{X}_0 \}. \quad (2)$$

If $\mathcal{X}_0$ carries a probability density p_0 , the propagated density is the pushforward $p_t = (\varphi_t)_\# p_0$, supported on $\mathcal{X}_t$. The goal of set-based propagation is a tractable, accurate representation of $\mathcal{X}_t$ (and, where needed, of p_t).

2.2 Differential algebra and the Taylor flow map

Parametrise the initial set by a vector of *factors* $\boldsymbol{\xi} \in [-1, 1]^p$ through a map $\mathbf{x}_0(\cdot) : [-1, 1]^p \rightarrow \mathbb{R}^n$, so that $\mathcal{X}_0 = \mathbf{x}_0([-1, 1]^p)$. Composing with the flow defines the *flow map*

$$\mathcal{T}_t(\boldsymbol{\xi}) = \varphi_t(\mathbf{x}_0(\boldsymbol{\xi})), \quad (3)$$

whose image is exactly $\mathcal{X}_t$. Differential algebra computes the order- N Taylor expansion of $\mathcal{T}_t$ about $\boldsymbol{\xi} = \mathbf{0}$ by carrying truncated polynomials through every arithmetic operation of the numerical integrator [4]. The result is a polynomial

$$\mathcal{P}_t(\boldsymbol{\xi}) = \sum_{|\boldsymbol{\alpha}| \leq N} \mathbf{c}_{\boldsymbol{\alpha}} \boldsymbol{\xi}^{\boldsymbol{\alpha}}, \quad \boldsymbol{\xi}^{\boldsymbol{\alpha}} = \prod_{k=1}^p \xi_k^{\alpha_k}, \quad (4)$$

with vector coefficients $\mathbf{c}_{\boldsymbol{\alpha}} \in \mathbb{R}^n$ indexed by multi-indices $\boldsymbol{\alpha} \in \mathbb{N}_0^p$. The truncation error $\mathcal{T}_t - \mathcal{P}_t$ is governed by the magnitude of the highest-retained-order terms and grows as the domain deforms under the flow [14]. Evaluating (4) at any $\boldsymbol{\xi} \in [-1, 1]^p$ predicts the propagated state of the corresponding initial condition without re-integration; this is the property that makes a single DA propagation stand in for many trajectories [13].

2.3 Test bed: the planar two-body problem

We use the planar Kepler problem in canonical units ($\mu = 1$, semi-major axis $a = 1$),

$$\ddot{\mathbf{r}} = -\frac{\mathbf{r}}{\|\mathbf{r}\|^3}, \quad \mathbf{r} = (x, y), \quad (5)$$

with state $\mathbf{x} = (x, y, v_x, v_y)$. The reference orbit has eccentricity $e = 0.5$ and is taken at periaapsis, $x(0) = a(1 - e) = 0.5$, $v_y(0) = \sqrt{(1 + e)/(1 - e)} = \sqrt{3}$, with the remaining components zero; its period is $T = 2\pi$. The two-body problem is an informative test bed because its nonlinearity is strong near periaapsis and mild near apoapsis, so the difficulty of propagation varies along a single orbit, and because exact references are available for validation. All DA propagations below use truncation order $N = 8$ and an adaptive Runge–Kutta integrator with tolerance 10^{-12} .

3 Set representations

We summarise the relevant representations, each describing a set as the image of the factor box $[-1, 1]^p$ under a map of increasing generality, and make precise the identification used throughout.

Definition 1 (Interval / box). *A box is the image of the factor cube under a diagonal affine map,*

$$\mathcal{B} = \{ \mathbf{c} + \text{diag}(\mathbf{h}) \boldsymbol{\xi} : \boldsymbol{\xi} \in [-1, 1]^p \}, \quad (6)$$

with centre $\mathbf{c}$ and half-widths $\mathbf{h}$. It can stretch only along the coordinate axes.

Definition 2 (Zonotope [6]). *A zonotope replaces the diagonal by a general generator matrix $G \in \mathbb{R}^{n \times p}$,*

$$\mathcal{Z} = \{ \mathbf{c} + G \boldsymbol{\xi} : \boldsymbol{\xi} \in [-1, 1]^p \} = \mathbf{c} \oplus \bigoplus_{j=1}^p [-1, 1] \mathbf{g}_j, \quad (7)$$

i.e. the Minkowski sum of the line segments spanned by the generator columns $\mathbf{g}_j$. When $p = n$ and G is nonsingular, $\mathcal{Z}$ is a parallelotope: an affine image of the cube, free to rotate and shear. A box is the special case $G = \text{diag}(\mathbf{h})$. Every zonotope is convex and centrally symmetric about $\mathbf{c}$.

Definition 3 (Polynomial zonotope [1, 8]). *A polynomial zonotope lets the factors enter through a polynomial,*

$$\mathcal{PZ} = \left\{ \mathbf{c} + \sum_{i=1}^h \left(\prod_{k=1}^p \xi_k^{E_{ki}} \right) \mathbf{g}_i : \boldsymbol{\xi} \in [-1, 1]^p \right\}, \quad (8)$$

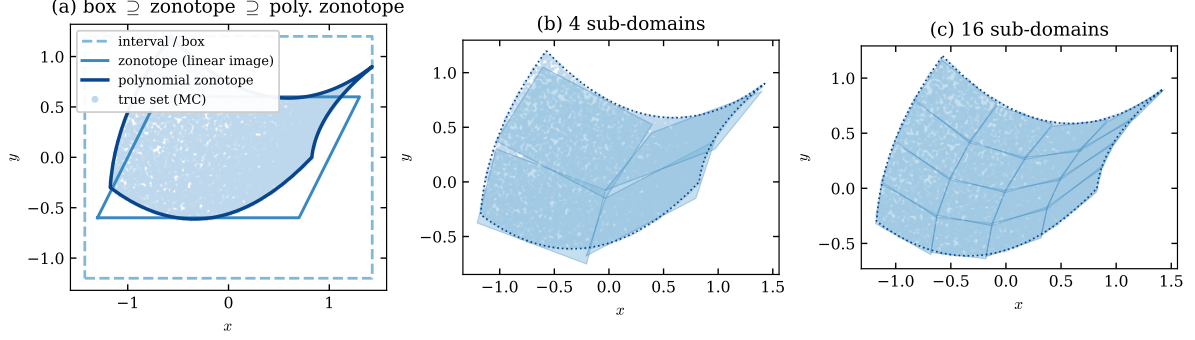


Figure 1: Set representations on a degree-two map. (a) The polynomial zonotope (darkest) coincides with the true set (pale points); the box (dashed) is a loose enclosure and the linear zonotope image misses the curvature. (b,c) Subdividing the factor box and taking each piece’s linear image converges onto the curved set—the principle behind ADS. All shades are drawn from the Blues colormap.

with an exponent matrix $E \in \mathbb{N}_0^{p \times h}$ and generators $\mathbf{g}_i \in \mathbb{R}^n$. Unlike a zonotope, a polynomial zonotope can be non-convex and asymmetric; it can bend and fold.

Comparing (4) and (8) gives the identification that organises this paper.

Proposition 1. *The truncated DA flow map $\mathcal{P}_t$ of (4), regarded as a set over $\boldsymbol{\xi} \in [-1, 1]^p$, is a polynomial zonotope: its centre is the constant term $\mathbf{c}_0$, its generators are the nonconstant coefficients $\mathbf{c}_\alpha$, and its exponent matrix collects the corresponding multi-indices α .*

Thus DA propagation produces a polynomial zonotope per domain, and the single-segment objects (6)–(7) are the degree-one special cases used to describe the *initial* set or to over-approximate the propagated one. Constrained zonotopes [12] and constrained polynomial zonotopes [9] extend the hierarchy with equality constraints $A\boldsymbol{\xi} = \mathbf{b}$ on the factors, which can represent arbitrary convex polytopes and lower-dimensional sub-manifolds; they are not used in the experiments below but are relevant to the applications discussed in Section 9.

Figure 1 illustrates the hierarchy on an explicit degree-two map $\varphi(p, q) = (p + \frac{1}{2}q + 0.35q^2, q + 0.45p^2 - 0.25pq)$ applied to a square of initial conditions. Because the map is degree two and $N \geq 2$, the polynomial zonotope coincides with the true image (the Monte Carlo cloud fills it exactly); the box is a loose enclosure, and the linear (zonotope) image—an affine map of the box—cannot reproduce the curvature. The right panels show that subdividing the factor box and taking the per-piece linear image converges onto the curved set: this is the geometric content of automatic domain splitting.

4 Automatic domain splitting

ADS makes the polynomial representation reliable for large sets by subdividing the factor domain [14]. Let $m_N(\mathcal{P}) = \sum_{|\alpha|=N} \|\mathbf{c}_\alpha\|_1$ denote the total magnitude of the highest-order coefficients, a proxy for the truncation error over $[-1, 1]^p$. The driver integrates the flow map and, at each accepted step, tests m_N against a tolerance ε . If $m_N > \varepsilon$ the domain is bisected along the factor ξ_d that contributes most of m_N ; otherwise the leaf is accepted.

Bisection along ξ_d replaces the factor by two half-ranges through the affine substitution

$$\xi_d \mapsto \pm \frac{1}{2} + \frac{1}{2} \xi'_d, \quad (9)$$

applied to the polynomial (4); expanding $(\pm \frac{1}{2} + \frac{1}{2} \xi'_d)^{\alpha_d}$ by the binomial theorem re-expresses the flow map in the child’s local coordinates without re-integration. Each child is again a polynomial zonotope on its sub-box, and the process recurses until every leaf satisfies the tolerance or a

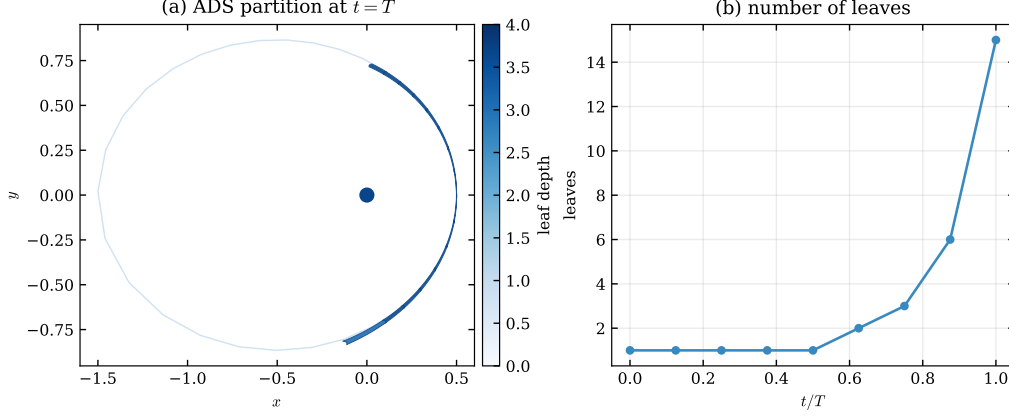


Figure 2: Automatic domain splitting on a two-body initial-condition box. (a) Leaves of the partition at $t = T$, shaded by subdivision depth; the partition refines where the dynamics are most nonlinear. (b) Number of leaves along the orbit.

maximum depth is reached. The output is a tree whose leaves tile the propagated set: a *piecewise polynomial zonotope*. The low-order variant LOADS replaces the truncation-mass criterion by a nonlinearity index built from the variation of the Jacobian, which can reduce the expansion order needed for a given accuracy [10].

Figure 2 shows ADS applied to a box of initial conditions on the two-body orbit, with leaves shaded by their subdivision depth. The partition is coarse where the flow is mild and fine where periapsis passage stretches the set; the leaf count grows accordingly along the orbit.

5 Oriented domains

The DA flow map and the substitution (9) operate entirely in the normalised factors ξ ; they are unaffected by how those factors map to physical initial states. That mapping—the geometry of the initial set—therefore can be chosen freely. The classical choice is the axis-aligned box (6). Replacing it by a parallelotope (7) with a general generator matrix G lets a single domain wrap a *correlated* or rotated initial set that a box would have to over-bound. Because subdivision bisects a generator (a column of G), the cuts follow the oriented directions of the set rather than the coordinate axes.

The effect is a smaller leaf count. Consider a two-body uncertainty in (y, v_y) shaped as a parallelogram rotated by 45° , and compare its propagation as an oriented zonotope against the axis-aligned box that bounds the same set. Over a representative arc the oriented domain needs fewer leaves at every snapshot (e.g. 48 versus 75 near mid-arc), because the bounding box covers empty corners that carry additional truncation mass and trigger spurious splits (Figure 3). The advantage is configuration dependent: a *fixed* orientation is a gamble, and over a full period the same 45° frame can become misaligned with the periapsis-return shear and lose to the box. This motivates choosing the orientation from the dynamics.

6 Flow-aligned domains

When the initial uncertainty is an *ellipsoid*—a covariance $\Sigma = LL^\top$ —the covering parallelotope is free to orient, because an ellipsoid has no preferred generator frame: for any orthogonal V , the parallelotope $\{\mathbf{c} + LV\xi : \xi \in [-1, 1]^p\}$ contains the same ellipsoid (the unit ball is contained in the unit cube, and V maps the cube to a set still containing the ball). This freedom can be spent to condition the *propagated* set.

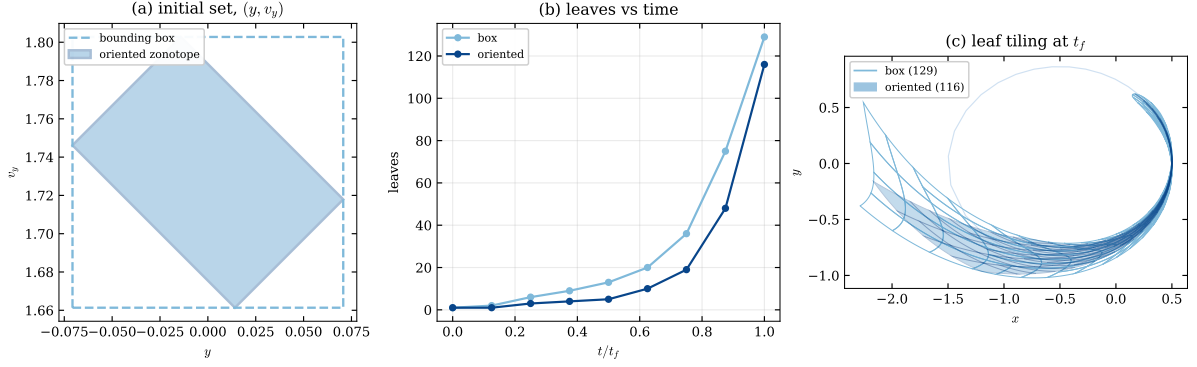


Figure 3: Oriented (parallelotope) versus axis-aligned (box) initial set on the two-body problem. (a) The correlated initial set and its bounding box in (y, v_y) . (b) Leaf count along the arc. (c) Leaf tiling at the final time; the box wastes leaves on the region its bound includes, while the oriented domain hugs the propagated set.

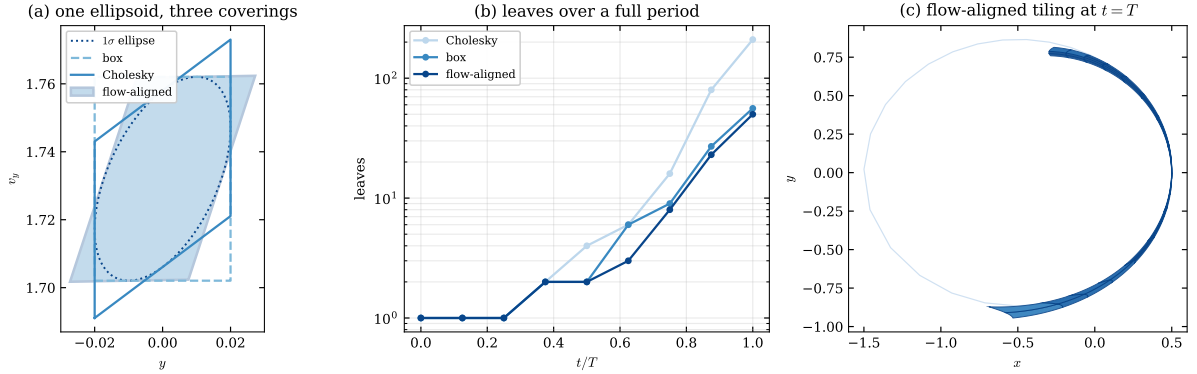


Figure 4: Flow-aligned covering of a covariance ellipsoid. (a) The 1σ ellipse and three parallelotope/box coverings in (y, v_y) . (b) Leaf count over a full period (log scale): the flow-aligned frame is lowest throughout, the misaligned Cholesky frame highest. (c) The flow-aligned partition at $t = T$, shaded by depth.

Let $\Phi = \partial\varphi_T/\partial\mathbf{x}_0$ be the state-transition matrix (the linear part of the flow map) over the horizon, obtained from a single un-split DA propagation. Take the thin singular value decomposition $\Phi L = U\Sigma_s V^\top$ and choose generators $G = LV$. Then the propagated parallelotope has linear image $\Phi G = U\Sigma_s$, whose columns are orthogonal and ordered by stretch: the set is mapped to a well-conditioned, axis-aligned (in the singular frame) parallelotope rather than a thin diagonal sliver that axis-aligned cuts would over-subdivide.

Figure 4 compares three coverings of the *same* ellipsoid over a full period: the axis-aligned bounding box, a fixed Cholesky parallelotope (oriented by the covariance only), and the flow-aligned parallelotope. The flow-aligned frame needs the fewest leaves at every snapshot (**50** versus **56** for the box and **210** for the Cholesky frame at $t = T$), *fixing* the periapsis-return penalty that a fixed orientation suffers. Notably the flow-aligned covering is *larger* in area than the box yet subdivides less: the gain is from orientation, not size, and a poorly oriented frame (the fixed Cholesky one) is several times worse. Re-expressing the propagated map in the aligned frame—a linear change of factor variables applied to the polynomial—confirms that the deformation is diagonalised: the off-diagonal of $\Phi^\top\Phi$ in the aligned frame drops from ≈ 1.2 to ≈ 0 .

A geometric caveat bounds how far the orientation can be pushed. A linear change of factor variables $\boldsymbol{\xi} = R\boldsymbol{\eta}$ maps the cube $[-1, 1]^p$ to $R[-1, 1]^p$, which is a cube again only when R is a signed permutation. Re-orienting the frame of a *fixed* parallelotope therefore either changes

the represented set or requires an over-approximation back to a cube. The construction above is exact precisely because an ellipsoid leaves the covering parallelotope free to orient; for a covariance the orientation is chosen once, up front, from the horizon Φ .

7 Curved initial sets

The representations so far take an affine *initial* geometry and let the flow introduce curvature. But the initial uncertainty itself is often curved. This is the generic situation when the uncertainty is Gaussian in one coordinate system and required in another: an admissible region in range/range-rate, a Gaussian in orbital elements expressed in Cartesian state, or a phase uncertainty along an orbit. We ask what a linear initial representation can and cannot do, and use a polynomial-zonotope initial set instead.

Why an affine set cannot represent a curved one. A linear initial set is $\mathbf{x}_0(\boldsymbol{\xi}) = \mathbf{c} + G\boldsymbol{\xi}$, an affine image of the box, hence a parallelotope: flat faces, straight edges, central symmetry about $\mathbf{c}$. That is the entire expressive range of a linear or covariance representation. Let the true initial map be $\mathbf{g} : [-1, 1]^p \rightarrow \mathbb{R}^n$, and expand it about the centre,

$$\mathbf{x}_0(\boldsymbol{\xi}) = \mathbf{g}(\mathbf{0}) + J\boldsymbol{\xi} + \frac{1}{2}(\boldsymbol{\xi}^\top H_i \boldsymbol{\xi})_{i=1}^n + \cdots, \quad (10)$$

with Jacobian J and component Hessians H_i . The best linear representation keeps only $\mathbf{c} + J\boldsymbol{\xi}$ and lies on the tangent plane at the centre; the true set departs from it by the quadratic term. Scaling the factors so that $\boldsymbol{\xi} = \pm \mathbf{1}$ corresponds to the 3σ boundary, the displacement is $\boldsymbol{\xi} \sim O(1)$, so the dropped curvature $\frac{1}{2}\boldsymbol{\xi}^\top H_i \boldsymbol{\xi}$ is the *same order* as the linear spread $J\boldsymbol{\xi}$ itself—not a small correction. No affine map produces a curved, asymmetric set; the deficiency is structural, not a matter of fitting G better. A polynomial zonotope retains the higher-order terms of (10) and bends to follow the manifold.

Construction. For the two-body problem, take the uncertainty to be a 3σ Gaussian in true anomaly ν and eccentricity e about $(\nu_0, e_0) = (0, 0.5)$, mapped to Cartesian state through the perifocal relations

$$r = \frac{a(1 - e^2)}{1 + e \cos \nu}, \quad \mathbf{r} = r(\cos \nu, \sin \nu), \quad \dot{\mathbf{r}} = \sqrt{\frac{\mu}{a(1 - e^2)}} (-\sin \nu, e + \cos \nu). \quad (11)$$

Substituting $\nu = 3\sigma_\nu \xi_1$ and $e = e_0 + 3\sigma_e \xi_2$ and expanding (11) as a DA polynomial yields a curved polynomial-zonotope initial set $\mathbf{x}_0(\boldsymbol{\xi})$, which is propagated by ADS exactly as before. The first-order truncation of $\mathbf{x}_0$ provides the linear (tangent) comparison, the parallelotope a covariance would supply.

Result. With $\sigma_\nu = 0.12$ (so the 3σ span is $\pm 21^\circ$) and $\sigma_e = 0.04$ ($e \in [0.38, 0.62]$), propagated to $0.75T$, the curved polynomial-zonotope set is represented by **18** leaves and reproduces the 10^4 -sample cloud with root-mean-square position error 2.6×10^{-3} (maximum 1.2×10^{-2}). The linear initial set, by contrast, requires **172** leaves and attains only RMS 0.48 (maximum 11): its flat 3σ geometry is wrong from the outset, and the nonlinear flow amplifies the error—indeed its tangent set reaches near-collision states that diverge if the horizon or the spread is pushed further. At 3σ the curvature is not a correction but the difference between a correct answer and a useless one (Figure 5).

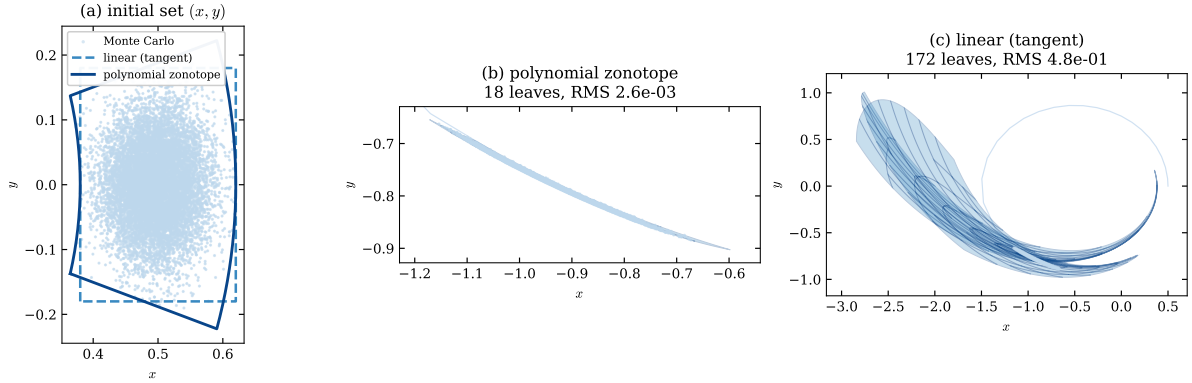


Figure 5: Curved initial set (Gaussian in orbital elements) versus its linear approximation. (a) Initial sets in (x, y) over the Monte Carlo cloud: the polynomial zonotope (solid) bends with the cloud, the tangent parallelotope (dashed) is flat. (b) Propagated, the curved set tracks the cloud with 18 leaves. (c) The linear set, propagated, smears far past the cloud (172 leaves).

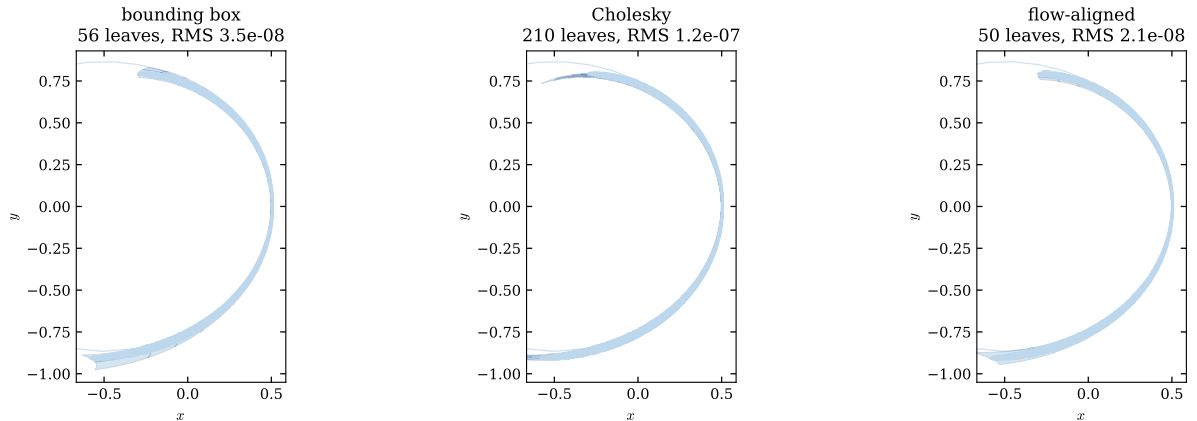


Figure 6: Monte Carlo validation for the covariance-ellipsoid case at $t = T$: the leaf tiling of each covering over the 10^4 -sample cloud (pale points). All three envelope the cloud; the leaf counts differ.

8 Monte Carlo validation

To confirm that the piecewise polynomial zonotope does not merely *bound* the propagated set but *reproduces* it, we draw 10^4 samples from each initial distribution, propagate them with a high-accuracy integrator, locate the leaf containing each sample, evaluate that leaf’s flow map, and measure the prediction error. Figure 6 overlays the partition on the cloud for the ellipsoidal case; the leaf tilings envelope the samples in every covering.

Table 1 collects leaf counts and errors. All coverings reproduce the cloud to the split tolerance, so the choice of domain geometry is a question of *cost* (leaf count) rather than achievable accuracy—except for the linear *initial* set, which cannot represent the curved geometry and is therefore inaccurate at any leaf count. The flow-aligned covering attains both the fewest leaves and the lowest error for the ellipsoidal case.

9 Discussion

Choosing a representation. The experiments suggest a simple decision rule. For a *filled, affine* uncertainty (a box or a covariance ellipsoid), the propagated set is recovered by ADS over a box; an *oriented* parallelotope reduces the leaf count when the set is correlated or anisotropic, and for an ellipsoid the orientation is best taken from the state-transition matrix, which is

Table 1: Leaf count and Monte Carlo prediction error (10^4 samples) for the domain geometries studied. Errors are position RMS / maximum over the samples. “Parallelotope” and “ellipsoid” refer to the initial set; “curved” is the polynomial-zonotope initial set of Section 7.

Initial set	Domain geometry	Leaves	RMS error	Max error
Parallelotope ($0.8T$)	bounding box	129	4.7×10^{-8}	5.2×10^{-7}
	oriented zonotope	116	1.1×10^{-7}	1.9×10^{-6}
Ellipsoid (T)	bounding box	56	3.5×10^{-8}	3.8×10^{-7}
	Cholesky frame	210	1.2×10^{-7}	3.4×10^{-6}
	flow-aligned	50	2.1×10^{-8}	2.5×10^{-7}
Curved ($0.75T$)	polynomial zonotope	18	2.6×10^{-3}	1.2×10^{-2}
	linear (tangent)	172	4.8×10^{-1}	1.1×10^1

exact and cheap. For a *curved* initial uncertainty—non-Gaussian in the working coordinates—a polynomial-zonotope initial set is required: no affine geometry can represent it, and using one produces both more leaves and a wrong answer.

On the 3σ ellipse. A filled 3σ ellipse is full-dimensional and needs no constraints; the oriented/flow-aligned covering of Section 6 handles it. Constraints [9, 12] become useful when the question concerns a *sub-manifold* of the ellipse rather than the whole set: its 3σ *shell* (a single quadratic constraint, the worst-case contour); a stiff direction of a rank-deficient covariance; or the sub-population that triggers an event at a later time, i.e. the preimage $\{\xi : \varphi_T(\mathbf{x}_0(\xi)) \in \mathcal{G}\}$ of a keep-out or close-approach region $\mathcal{G}$, which pulls back through the leaf polynomial to a polynomial constraint on the factors. Because the constraint lowers the dimension, it both reduces the number of sub-domains and sharpens the answer; the collision probability is then an integral of the initial Gaussian over the constrained factor region rather than a Monte Carlo hit count. These are natural extensions of the present work.

Cost and limitations. The cost of a curved (polynomial-zonotope) domain is that each leaf carries its initial map in addition to the propagated one, roughly doubling per-leaf memory, and exact point location requires inverting a polynomial map (we use a first-order inverse on the small leaves). The flow-aligned construction assumes an ellipsoidal initial set and a single horizon-level orientation; time-varying re-orientation would require an over-approximation, as noted in Section 6. Finally, the present study is two-dimensional in the uncertain factors and uses a single dynamical model; the representations carry over to higher factor dimension and to perturbed dynamics without change, but the leaf-count scaling with factor dimension—the curse that ADS shares with all grid-like subdivisions—remains the practical limit.

10 Conclusion

We identified the DA flow map propagated by automatic domain splitting with the polynomial zonotope of reachability analysis, and used that identification to study three usage patterns on the two-body problem: oriented domains for correlated sets, flow-aligned domains for covariance ellipsoids, and curved polynomial-zonotope initial sets for non-Gaussian uncertainty. Validated against 10^4 Monte Carlo samples, the combined approach reproduces the propagated set to tolerance while using markedly fewer sub-domains than an axis-aligned box, and the curved representation captures initial uncertainty that no affine set can. The set-representation hierarchy developed for formal verification and the adaptive, error-driven subdivision developed for astrodynamics are complementary, and their combination is a practical tool for nonlinear orbital-uncertainty propagation.

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